Feature-Based Clustering of Time Series Using Ordinal Pattern Analysis

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08 December 2025

Outline

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Motivation and Background

- Time series clustering is challenging since every series analyzed contains a stochastic component, contributing random variability and increased complexity.
- Ordinal patterns capture the order relations among consecutive data points.
- **Bandt & Pompe (2002)** [**?**]: introduced **Permutation Entropy**.
- **Limitation:** Entropy quantifies disorder but cannot distinguish random from structured dynamics.
- **Extension:** Statistical Complexity.
- The **Entropy–Complexity Plane** maps time series by entropy (H) and complexity (C).
- **Goal:** Differentiate AR, MA, and ARMA models.

From Time Series to Ordinal Patterns: An Example

- **Example:** Mean monthly humidity in Wellington.
- Original time series:

$$77.3,\ 81.0,\ 82.4,\ 81.7,\ 83.6,\ 85.6,\ \ldots$$

Figure: Mean monthly humidity in Wellington

Transforming into Ordinal Patterns

- Choose embedding dimension $D = 3$.
- Slide a window of size D across the series:

$$(77.3, 81.0, 82.4) \to (1, 2, 3) = \pi_1$$
$$(81.0, 82.4, 81.7) \to (1, 3, 2) = \pi_2$$

- Continue for all overlapping windows ($n - D + 1$ total).
- Each window represents an ordinal pattern describing local order relations.

Definition: Permutation Entropy (H)

$$H(\mathbf{p}) = -\frac{1}{\log k}\sum_{i=1}^{k} p(\pi^i)\log p(\pi^i),$$

where $k = D!$ and $p(\pi^i)$ is the proportion of pattern π^i.

Definition: Jensen-Shannon Distance

Given a histogram $\mathbf{p}$ and the reference uniform distribution $\mathbf{u} = (1/k, \ldots, 1/k)$, the disequilibrium is calculated as:

$$Q'(\mathbf{p},\mathbf{u}) = \sum_{\ell=1}^{k}\left(p_\ell \log\frac{p_\ell}{u_\ell} + u_\ell \log\frac{u_\ell}{p_\ell}\right).$$

Disequilibrium and Statistical Complexity

Disequilibrium

The **normalized disequilibrium**:

$$Q = \frac{Q'}{\max(Q')},$$

provides a scale-independent comparison, where $\max(Q')$ is the highest possible value for the given k.

Definition: Statistical Complexity (C)

$$C = HQ$$

where H is normalized entropy and Q is normalized disequilibrium.

Conceptual Framework

- Transform time series $\{x_t\}$ into **ordinal patterns**.
- Compute:
  - Permutation Entropy
  - Statistical Complexity
  - Confidence Interval for permutation entropy and complexity
- Plot in the **Entropy–Complexity** ($H \times C$) plane.
- Use cluster formation to discriminate models.

Figure: Entropy–Complexity plane for Sample size $n = 1000$.

Experimental Design

- Models studied:
  - AR(1), AR(2), MA(1), MA(2), ARMA(1,1), ARMA(2,2)
- Each model simulated with 100 independent replications.
- Parameter sets chosen to ensure stationarity and invertibility.
- Series lengths: $n = 1000, n = 5000$.

Model Parameter Structure

Type	Model	α_1	α_2	β_1	β_2
AR(1)	$\mathrm{AR}(1)_{\mathrm{M}_1}$	0.8			
	$\mathrm{AR}(1)_{\mathrm{M}_2}$	0.1			
MA(1)	$\mathrm{MA}(1)_{\mathrm{M}_1}$			0.8	
	$\mathrm{MA}(1)_{\mathrm{M}_2}$			0.1	
ARMA(1,1)	$\mathrm{ARMA}(1,1)_{\mathrm{M}_1}$	0.8		0.8	
	$\mathrm{ARMA}(1,1)_{\mathrm{M}_2}$	0.1		0.1	

Table: Example of parameter sets ensuring diversity.

Feature Extraction and Central Points

- For each realization, compute:
  - Entropy (H) — randomness
  - Complexity (C) — structure
  - Confidence Interval(CI) for H and C
- For each model, identify the **central point** (H^*, C^*) using PCA projection.
- Central ensures the point is the most representative of the model's behavior, as referred to in Chagas et al. [**?**].

Case Study Design

- **Case 1–3:** Positive, negative, and mixed-sign coefficients.
- Two sample sizes: $n = 1000, n = 5000$.
- Each case analyzed using:
  - The density of points in the $H \times C$
  - Scatter plots with confidence intervals for H and C
  - Heat maps for model classification

Case 1 — Positive Coefficients

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Figure: Entropy–complexity scatter plot ($n = 1000$ and $n = 5000$) for positive model coefficients.

Case 1 — Positive Coefficients cont. . .

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Figure: Entropy–complexity central points and associated time series ($n = 5000$) and heat map for positive model coefficients.

Case 2 — Negative Coefficients

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Figure: Entropy–complexity scatter plot ($n = 1000$ and $n = 5000$) for negative model coefficients.

Case 2 — Negative Coefficients cont. . .

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Figure: Entropy–complexity central points and associated time series ($n = 5000$) and heat map for negative model coefficients.

Case 3 — mixed-Sign Coefficients

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Figure: Entropy–complexity scatter plot ($n = 1000$ and $n = 5000$) for mixed model coefficients.

Case 3 — Mixed-Sign Coefficients cont. . .

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Figure: Entropy–complexity central points and associated time series ($n = 5000$) and heat map for mixed model coefficients.

Combined Model Visualization

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Key Findings

- Ordinal pattern analysis successfully distinguishes AR, MA, and ARMA models.
- Distinct clustering appears across Permutation entropy.
- Central (H^*, C^*) points provide consistent model representation.
- Larger samples yield tighter clusters and improved discrimination.

Conclusion

Summary

This study reveals that combining ordinal pattern analysis with entropy–complexity measures provides a robust framework for model discrimination in simulated time series.

- Captures both linear and nonlinear relationship between the entropy and complexity.
- Overlap points reveal intrinsic similarities in model dynamics.
- Extensible to real-world applications such as finance, engineering, biomedical signals.

Future Work

- Apply unsupervised clustering (e.g., k-means, hierarchical) in the H–C plane.
- Extend analysis by increasing sample size
- Extend distinct clustering using other types of entropies Rényi, Tsallis, and Fisher Information.
- Extend analysis to empirical systems.

Thank You!

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References I